CHAPTER 4

DETERMINANTS

CHAPTER ESSENCE AT A GLANCE:
Determinant evaluation, area of triangles, minors, cofactors, adjoint of a matrix, matrix inversion, and solving linear systems AX = B.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Determinant as a Scalar Measure:** A determinant associates a single unique scalar value with a square matrix, evaluated through cofactor expansion along any chosen row or column where rows with maximal zeros yield optimal computation speed.

• **Invertibility and Adjoint Formulation:** A square matrix possesses a multiplicative inverse if and only if its determinant is non-zero, with the inverse explicitly computed as the adjoint matrix divided by the determinant scalar.

• **Coordinate Area and Collinearity:** The geometric area of a triangle formed by three coordinate vertices in a plane is given by half the absolute value of their coordinate determinant, vanishing to zero precisely when the points lie along a single straight line.

• **Matrix Method for Linear Systems:** A system of linear equations represented in matrix form AX equals B has a unique consistent solution when the coefficient determinant is non-zero, evaluated directly by multiplying the inverse of matrix A by constant matrix B.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$A^{-1} = \frac{1}{|A|}\text{adj}(A) \quad (|A| \neq 0)$$

Matrix inverse by adjoint method for non-singular matrices

$$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = |A|I$$

Fundamental identity relating matrix, adjoint, and determinant

$$|\text{adj}(A)| = |A|^{n-1} \quad \text{and} \quad |kA| = k^n|A|$$

Determinant of adjoint matrix and scalar scaling for order n

$$\Delta = \tfrac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

Area of triangle formed by three vertices in Cartesian plane

$$AX = B \rightarrow X = A^{-1}B = \frac{1}{|A|}\text{adj}(A)B$$

Matrix method solution for system of linear equations

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Area Sign] When area is given as 4 sq. units, write determinant = ±4! Omitting the negative sign loses 2 marks by missing the second value of k.

• **Alert:** [Trap Alert: Adjoint Transpose] In finding adj(A), remember to TRANSPOSE the cofactor matrix! Putting cofactors directly without transpose is the #1 student error.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Evaluate the determinant of order 3: |[[3, -1, -2], [0, 0, -1], [3, -5, 0]]|. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Evaluation along Row 2 (containing two zeros)]** |A| = |[[3, -1, -2], [0, 0, -1], [3, -5, 0]]| Expanding along R_2: |A| = - 0 · M_{21} + 0 · M_{22} - (-1) · |[[3, -1], [3, -5]]| = + 1 · [3(-5) - (-1)(3)] = + 1 · [-15 - (-3)] = -15 + 3 = -12. [2 Marks]

• **[Boxed Result: Determinant Value = -12]**

2. Find the area of triangle with vertices at (3, 8), (-4, 2), and (5, 1) using determinants. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Area of Triangle Formula using Determinants]** Δ = (1 over 2) | |[[x_1, y_1, 1], [x_2, y_2, 1], [x_3, y_3, 1]]| |. [0.5 Mark]

• **[Calculation]** Vertices (3, 8), (-4, 2), (5, 1): D = |[[3, 8, 1], [-4, 2, 1], [5, 1, 1]]| = 3(2(1) - 1(1)) - 8(-4(1) - 1(5)) + 1(-4(1) - 2(5)) = 3(2 - 1) - 8(-4 - 5) + 1(-4 - 10) = 3(1) - 8(-9) + 1(-14) = 3 + 72 - 14 = 61. Area Δ = (1 over 2) |61| = 61 over 2 = 30.5 sq. units. [1.5 Marks]

• **[Boxed Result: Area of Triangle = 61 over 2 = 30.5 sq. units]**

3. For matrix A = [[2, 3], [1, 4]], find adj(A) and verify A(adj A) = (adj A)A = |A| I. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Compute adj(A) and |A|]** A = [[2, 3], [1, 4]]. |A| = 2(4) - 3(1) = 8 - 3 = 5. Cofactors: A_{11} = +4, A_{12} = -1, A_{21} = -3, A_{22} = +2. adj(A) = [[A_{11}, A_{21}], [A_{12}, A_{22}]] = [[4, -3], [-1, 2]] (swap diagonals, negate off-diagonals). [1.5 Marks]

• **[Step 2: Verification of A(adj A)]** A · adj(A) = [[2, 3], [1, 4]] [[4, -3], [-1, 2]] = [[2(4)+3(-1), 2(-3)+3(2)], [1(4)+4(-1), 1(-3)+4(2)]] = [[5, 0], [0, 5]] = 5 [[1, 0], [0, 1]] = |A| I. adj(A) · A = [[4, -3], [-1, 2]] [[2, 3], [1, 4]] = [[4(2)-3(1), 4(3)-3(4)], [-1(2)+2(1), -1(3)+2(4)]] = [[5, 0], [0, 5]] = |A| I. Verified. [1.5 Marks]

• **[Boxed Result: A(adj A) = (adj A)A = |A| I = [[5, 0], [0, 5]]]**

4. Find the inverse of matrix A = [[2, -2], [4, 3]] using adjoint method. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Determinant & Adjoint]** A = [[2, -2], [4, 3]]. |A| = 2(3) - (-2)(4) = 6 + 8 = 14 ≠ 0 (Invertible). adj(A) = [[3, 2], [-4, 2]]. [1.5 Marks]

• **[Step 2: Inverse Calculation]** A^{-1} = (1 over |A|) adj(A) = (1 over 14) [[3, 2], [-4, 2]] = [[3 over 14, 1 over 7], [-2 over 7, 1 over 7]]. [1.5 Marks]

• **[Boxed Result: A^{-1} = (1 over 14) [[3, 2], [-4, 2]]]**

5. Find the inverse of matrix A = [[1, 2, 3], [0, 2, 4], [0, 0, 5]]. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Determinant of Upper Triangular Matrix]** A = [[1, 2, 3], [0, 2, 4], [0, 0, 5]]. |A| = 1 × 2 × 5 = 10 ≠ 0. [1 Mark]

• **[Step 2: Cofactors & Adjoint]** A_{11} = 10, A_{12} = 0, A_{13} = 0. A_{21} = -10, A_{22} = 5, A_{23} = 0. A_{31} = 2, A_{32} = -4, A_{33} = 2. adj(A) = [[10, -10, 2], [0, 5, -4], [0, 0, 2]]. [1 Mark]

• **[Step 3: Inverse]** A^{-1} = (1 over 10) [[10, -10, 2], [0, 5, -4], [0, 0, 2]] = [[1, -1, 1 over 5], [0, 1 over 2, -2 over 5], [0, 0, 1 over 5]]. [1 Mark]

• **[Boxed Result: A^{-1} = (1 over 10) [[10, -10, 2], [0, 5, -4], [0, 0, 2]]]**

6. Solve the system of equations by matrix method: 2x + 5y = 1, 3x + 2y = 7. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Matrix Form AX = B]** [[2, 5], [3, 2]] [[x], [y]] = [[1], [7]]. A = [[2, 5], [3, 2]], X = [[x], [y]], B = [[1], [7]]. [1 Mark]

• **[Inverting A]** |A| = 2(2) - 5(3) = 4 - 15 = -11 ≠ 0. adj(A) = [[2, -5], [-3, 2]]. A^{-1} = - (1 over 11) [[2, -5], [-3, 2]]. [1.5 Marks]

• **[Solving X = A^{-1} B]** X = - (1 over 11) [[2, -5], [-3, 2]] [[1], [7]] = - (1 over 11) [[2(1) - 5(7)], [-3(1) + 2(7)]] = - (1 over 11) [[2 - 35], [-3 + 14]] = - (1 over 11) [[-33], [11]] = [[3], [-1]]. Therefore, x = 3, y = -1. [1.5 Marks]

• **[Boxed Result: x = 3, y = -1]**

7. Solve the system of linear equations by matrix method: x - y + 2z = 7, 3x + 4y - 5z = -5, 2x - y + 3z = 12.
[Score: 5 Marks]

Answer & Key Valuation Steps:

• **[System in Matrix Form AX = B]** A = [[1, -1, 2], [3, 4, -5], [2, -1, 3]], X = [[x], [y], [z]], B = [[7], [-5], [12]]. [1 Mark]

• **[Determinant |A|]** |A| = 1(12 - 5) - (-1)(9 - (-10)) + 2(-3 - 8) = 1(7) + 1(19) + 2(-11) = 7 + 19 - 22 = 4 ≠ 0. [1 Mark]

• **[Cofactors & Adjoint]** A_{11} = 7, A_{12} = -19, A_{13} = -11. A_{21} = 1, A_{22} = -1, A_{23} = -1. A_{31} = -3, A_{32} = 11, A_{33} = 7. adj(A) = [[7, 1, -3], [-19, -1, 11], [-11, -1, 7]]. [1.5 Marks]

• **[Solution $X = A^{-1} B$]** X = (1 over 4) [[7, 1, -3], [-19, -1, 11], [-11, -1, 7]] [[7], [-5], [12]] = (1 over 4) [[49 - 5 - 36], [-133 + 5 + 132], [-77 + 5 + 84]] = (1 over 4) [[8], [4], [12]] = [[2], [1], [3]]. Hence x = 2, y = 1, z = 3. [1.5 Marks]

• **[Boxed Result: x = 2, y = 1, z = 3]**